$$\mathcal{L}_{\text{dice}} = 1 - \frac{2\sum_i p_i y_i + \epsilon}{\sum_i p_i + \sum_i y_i + \epsilon}$$

$$\mathcal{L} = 20\mathcal{L}_{\text{focal}} + \mathcal{L}_{\text{dice}} + \mathcal{L}_{\text{IoU}} + \mathcal{L}_{\text{object}}$$